Highlights

The Boussinesq–compressible gap in two-dimensional steady Rayleigh–Bénard convection is linear to leading order in the relative temperature contrast

- Boussinesq–compressible gap follows $R = A\varepsilon^m$ with $m = 1.02$ in four channels
- Relative temperature contrast is the only axis; a $24\times$ Ma change is unmeasurable
- 627 paired solves fix the law; public data reproduce only its amplitude
- At $\varepsilon = 1.2$ integral Nu shifts 0.6% but local wall flux redistributes 4.25%
- Factored prior $D = \varepsilon P(X)$: $1.6\times$ over plain and $1.9\times$ over ε-blind networks